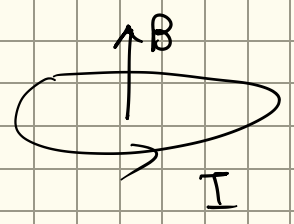
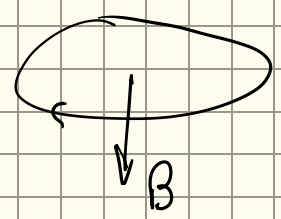


current counter clockwise

Reflecting

current clockwise



Note that if $A \in O(3)$ then $E \to A \cdot E$
$B \to (\det A) A \cdot B$

Exercise: Since $E = -\nabla V$ and $B = \nabla \times A$ show this.

Note we have a representation $A \mapsto \operatorname{sgn}(\det A) A$ of $GL(n, \mathbb{R})$.

Therefore this gives rise to a new vector bundle over M. This is the bundle of psuedovectors over M.

In $\mathbb{R}^3$ there is a correspondence between psudovectors and two forms.

let $P(x,y,z) = (-x,-y,-z)$ then
$$P^*(dx \wedge dy) = dx \wedge dy$$
$$P^*(dz \wedge dx) = dz \wedge dx$$
$$P^*(dy \wedge dz) = dy \wedge dz$$

we know $P^* B = -B \Rightarrow$ that components of B pick up a minus sign.

Hence if $B = B_1 \partial_x + B_2 \partial_y + B_3 \partial_z$ it natural to think of B as a 2 form
$\hat{B} = B_1\, dy \wedge dz + B_2\, dz \wedge dx + B_3\, dx \wedge dz$.

why? Because if $\alpha = \alpha^1 dx \wedge dy + \alpha^2 dz \wedge dx + \alpha^3 dy \wedge dz$
then $P^* \alpha = (\alpha^1 \circ P)\, dx \wedge dy + (\alpha^2 \circ P)\, dz \wedge dx + (\alpha^3 \circ P)\, dy \wedge dz$

Recall: if V is a finite dimensional vector space then $\dim \Lambda^k V = \dim \Lambda^{n-k} V$
$\binom{n}{k} = \binom{n}{n-k}$

We then defined $\star : \Lambda^k V \to \Lambda^{n-k} V$ s.t. $w \wedge \star v = \langle w, v \rangle \, vol_g$
$v \mapsto \star v$

let (M, g) be a (Pseudo) Riemannian manifold, $g \in \Gamma(\Sigma^2 T^* M)$ s.t. $\forall p \in M$ $g_p : T_pM \times T_pM \to \mathbb{R}$ is non degenerate.

we can extend $\star$ to $\Lambda^k T_p^* M \to \Lambda^{n-k} T_p^* M$ by $\eta \wedge \star \omega = \langle \eta, \omega \rangle dV_g$

we can extend this pointwise by: if $\omega \in \Omega^k(M)$ then $\star\omega \in \Omega^{n-k}(M)$ s.t.
$(\eta \wedge \star\omega)_p = \langle \eta, \omega \rangle_p \, dV_g|_p$

If M is compact we can define an extension of g_p to $\Omega^k(M)$ given by $(\omega, \eta) = \int_M \langle \omega, \eta \rangle dV_g$

$$= \int_M \omega \wedge \star \eta$$

$$= \int_M \eta \wedge \star \omega$$

Aside: If $T : V \to V$ where V is an (finite dimensional) inner product space then $T^* : V \to V$ s.t. $\langle Tv, w \rangle = \langle v, T^* w \rangle$ this exists due to Riesz representation.

let $d : \Omega^k(M) \to \Omega^{k+1}(M)$ [Note we can take $d : \underset{\bigoplus \Omega^k(M)}{\underset{\parallel}{\Omega^\bullet(M)}} \to \Omega^\bullet(M)$]

Then define $d^* : \Omega^{k+1}(M) \to \Omega^k(M)$ s.t. $(\omega, d\eta) = (d^*\omega, \eta)$.

Exercise: Show that $d^* = (-1)^{n(k+1)+(s+1)} \star d \star$ where s is the signature of g.

Note d^* is well defined on all of $\Omega^\bullet(M)$.

Let $\omega \in \Omega^k(M)$ and $\eta \in \Omega^{k+2}(M)$ then $(\omega, d^*d^*\eta) = (d\omega, d^*\eta)$
$= (d^2\omega, \eta)$
$= 0$

Note this says that $d^*d^*\eta = 0 \Rightarrow (d^*)^2 = 0$ [Take this lightly for now as I'm not too sure if $(\cdot,\cdot)$ is nondegenerate for nondegenerate g. But I think its true since its true in Riemannian]

$d + d^* : \Omega^k(M) \to \Omega^{k-1}(M) \oplus \Omega^{k+1}(M)$

Note that $(d+d^*)^2 = dd^* + d^*d = \Delta : \Omega^k(M) \to \Omega^k(M)$

We call ω harmonic if $\Delta\omega = 0$

Exercise: 1. On $\mathbb{R}^n$ this laplacian is the same as usual laplacian.

2. ω is harmonic $\Leftrightarrow$ $d\omega$ and $d^*\omega$ are null. (Norm 0)

Let (M, g) be a Lorentzian manifold. Suppose for now that $M = \mathbb{R} \times \mathbb{R}^3$ with $g = -dt^2 + dx^2 + dy^2 + dz^2$

Then let $E \in \Omega^1(M)$ and $B \in \Omega^2(M)$. Define $F = B + E \wedge dt$ then

This is just the stress energy tensor

$$\begin{aligned} dF &= dB + d(E \wedge dt) \\ &= dB + dE \wedge dt \\ &= d_S B + \partial_t B \wedge dt + d_S E \wedge dt \\ &= d_S B + (\partial_t B + d_S E) \wedge dt \end{aligned}$$

where d_S is the exterior derivative restricted to $\mathbb{R}^3 \subseteq M$ and $(\partial_t B)_{\mu\nu} = \partial_t B_{\mu\nu}$

Maxwell's Equations tell us that $\nabla \cdot B = 0 \Rightarrow d_S B = 0$
$\nabla \times E = -\frac{\partial B}{\partial t} \Rightarrow d_S E + \partial_t B = 0$

Exercise: Show that if $E \in \Omega^1(M)$ then $\nabla \cdot E = \star_S d_S \star_S E$ where $\star_S, d_S$ are hodge star on $\mathbb{R}^3$

Moreover show if $B \in \Omega^2(M)$ then $\nabla \times B = \star_S d_S \star_S B$.

What we can show is that $d^*F = J$ where $J = \rho\, dt + j^1 dx + j^2 dy + j^3 dz$ with ρ being the charge density and (j^1, j^2, j^3) being the current density.

In the language of differential forms $dF = 0$, $d^*F = J$ are the maxwells equations.

Yang Mills: $\mathcal{L} = \frac{1}{4} \operatorname{tr}(F \wedge \star F)$